

**DEVELOPING SUSTAINABLE INDUSTRIALIZATION POLICIES TO SUPPORT
INDONESIA'S G20 PRECIDENCY PRIORITIES: ENERGY TRANSITION**

Executive Summary

Energy transition is one of the priority sectors under Indonesia's G20 Presidency, which conveys the theme "Recover Together, Recover Stronger" in its spirit. As detailed in the leader's declaration, Indonesia must develop a comprehensive national energy reform strategy that addresses every sector, including industry, because industry plays a vital role in the economic development of a nation. Industrial policy is one of the primary drivers of industrialization; therefore, policies must react to environmental change, as the industrial sector is the most energy-intensive sector. To support this, the nation's industrial policy must incorporate the concept of sustainable industrialization, which ensures that economic development is accompanied by a significant reduction in emissions.

The National Industry Development Master Plan for the period 2015–2035 is governed by Government Regulation No. 14/2015 in Indonesia. This national industry development appears to be an ideal plan based on the strategies and objectives, but it is challenging to implement for the following reasons: 1) it requires cooperation from numerous stakeholders because each strategy is run by a different technical ministry (ego sectoral); 2) the goal of industrial sectors is not clear because every sector (from mainstream to upstream) has a priority; 3) Indonesia is experiencing negative deindustrialization; and 4) this plan does not contain the effort to reduce emissions and needs the adoption of the sustainable industrialization.

Several policy recommendations must be implemented to support the implementation of the sustainable industrialization concept in the National Industry Development Master Plan. 1) promoting carbon pricing for industry and environmental taxation: carbon pricing is an effective and inexpensive strategy for combating climate change Governments can support the development and adoption of green technologies, such as renewable energy and energy-efficient equipment, in the industrial sector. 3) implementing the circular economy – industrial symbiosis korean model: Korea has been successful in implementing this concept and has gained much benefit.

I. Introduction to Indonesia's G20 Presidency

The Indonesia's G20 Presidency carries the spirit of recovering together with the theme "Recover Together, Recover Stronger." Indonesia chose this theme since the world is currently under pressure due to the COVID-19 pandemic, requiring joint and inclusive efforts in finding a way out or solutions for recovery. According to the Ministry of Foreign Affairs of The Republic of Indonesia (2022) to accomplish this objective, the Indonesian Presidency is emphasizing on three priority sectors as the key to a robust and sustainable recovery, namely:

1. Strengthening of the Global Health Architecture

The current pandemic demonstrates that the global health architecture must be strengthened. This is necessary not only to combat the current pandemic, but also to better prepare the world to respond to future health crises. Threats to global health necessitate an effective multilateral and multisectoral response. With financial support, the G20 should focus on pandemic prevention, preparedness, and response (PPR).

2. Digital Transformation

The digital transformation has become a new source of economic development and is one of the important solutions for reviving the economy during a pandemic. To ensure an inclusive digital transformation for all, the Indonesian presidency will concentrate on enhancing digital skills and digital literacy. It is crucial that the G20 advances the global conversation about digital transformation in a fair manner.

3. Energy Transition

Renewable energy is a straightforward key performance indicator for monitoring the world's progress towards carbon neutrality. In order to promote the transition to renewable and sustainable energy, the Indonesian Presidency will prioritize energy security, accessibility, and affordability. This will assure a sustainable future and aid in the effective management of climate change issues, particularly in the industrial sector. The G20 plays a crucial role in catalyzing the transition to align public and private financial flows with the achievement of net-zero emissions.

The implementation of the Indonesian G20 Presidency is not merely a narrative or a written agreement. It is also tangible and provides Indonesia and the rest of the world actual, practical benefits. Indonesia also seeks to coordinate global policies that contribute to more balanced global governance, make the G20 more adaptable in the face of crises, and advocate for national interests in global forums.

II. Bali Leader's Declaration regarding Energy Transition

The G20 Secretariat has released a report on the Indonesian G20 Presidency's Leader's declaration. There are several Leader's Declaration regarding energy transition:

1. Ensure Global Energy Balances

The G20 is facing difficult problems from the energy and climate change crises, which are exacerbated by geopolitical issues (the Russia-Ukraine War). The world is experiencing volatile energy prices and markets, as well as energy supply shortages and disruptions. The G20 members commit to assuring that the global energy demand is met by affordable energy supplies with efforts to:

- a. Rapidly transform and diversify energy systems;
- b. Advance energy security and resilience, and market stability;
- c. Accelerate and ensure clean, sustainable, affordable, and inclusive energy transitions and the flow of sustainable investments; and
- d. Strengthen international cooperation on ensuring energy affordability and accessibility by limiting energy price volatility.

2. Providing Support for Developing Countries

The G20 members commit to continuing their support for developing countries, particularly the most vulnerable ones, in terms of providing access to affordable, reliable, sustainable, and modern energy, capacity building, affordable access to the latest technology in the public domain, mutually beneficial technology cooperation, and financing of energy sector mitigation actions.

3. Bali Energy Transition Roadmap (BETR)

The G20 countries' transition to low- and zero-emission development is intended to be accelerated by 2030 according to the Bali Energy Transition Roadmap. Securing energy accessibility, boosting smart and clean energy technologies, and expanding clean energy financing are The Bali Roadmap's three primary targets for quickening the energy transition. Numerous initiatives involving all stakeholders, from the government to the community, promote these three themes. Strengthening the government's approach to the national clean energy transition via planning, implementation, and tracking in energy policy decision-making is one of the activities that Indonesia should assist by enacting policy. Every sector in Indonesia should undergo a national energy transformation, including industry, which is the most energy-intensive sector.

4. Investment in Sustainable Infrastructure and Industry

The G20 members commit to promoting investment in sustainable infrastructure and industry, innovative technologies, and a broad range of fiscal, market, and regulatory mechanisms to support clean energy transitions, including, as appropriate, the use of carbon pricing and non-pricing mechanisms and incentives, while providing targeted support for the poorest and most vulnerable. Regarding this, G20 members should support sustainable industrialization through the approach of energy transition.

Regarding the energy transition in the industrial sector, there are two noteworthy pledges. Indonesia must first develop a comprehensive national energy reform strategy that addresses every sector, including industry. Second, encourage investment in environmentally friendly businesses. Both promises must be fulfilled through regulations. The government could demonstrate its commitment to promoting energy transition through regulation.

III. Industrialization Overview

There is widespread agreement that industrialization plays a crucial role in a nation's economic development, and there are numerous reasons why pursuing sustained industrialization has long-term benefits for economic development (Haraguchi et al. 2019). Some of these reasons are still rooted in Kaldor's law, which provides a conceptual framework for the link between manufacturing and economic growth. According to Mercan (2015), in Kaldor's first law, growth rate of the economy (GDP) is positively correlated with growth rate of the manufacturing industry sector. Kaldor identifies the manufacturing sector as a engine of growth. In its role as an economic catalyst, industrialization is affected by a few factors, with government policy being one of the most significant. In addition, industrialization is closely associated with the ensuing environmental impact.

III.1 Drivers of Industrialization

Industrialization is one of the pillars of sustainable economic expansion. As a result of the manufacturing sector's capacity to absorb labor force, increase diversification, and endure structural transformations while stimulating the development of other sectors, the manufacturing sector is a key driver of economic expansion (Haraguchi et al., 2019). According to Haraguchi et al. (2019), successful industrialization is driven by a combination of factors, including:

- a. Country's initial economic conditions: factor endowments and other characteristics, such as demography and geography
- b. High institutional stability: which in turn contributes to the creation of a sound investment climate.
- c. Economic policy and reform: policies in trade, capital flows and financial sector can largely affect industrial development.
- d. Government responsibility: policy makers play a crucial role in developing policies that are effective under different circumstances and in different regime.

UNIDO (2020) emphasizes that industrial policies have been a significant driver of industrialization in all successful industrialization experiences. Industrial policy has been utilized to cultivate social capability, direct market forces, stimulate technological and organizational innovation, and establish new markets and institutions.

III.2 Industrial Policy Adjustment

As previously stated, industrial policy is a significant generator of industrialization; therefore, industrial policies must adapt to context and time-specific country conditions, such as a nation's endowment structure, political economy, and government capabilities. According to UNIDO (2020), this is necessary because some policies that were feasible in the past may become unfeasible due to changes in global and national conditions. Additionally, some policies that were effective under a particular industrial and technological environment may need to be adapted to the changing industrial and technological environment and updated. Therefore, the government should learn how to resolve these obstacles so that industrial policy can be utilized more effectively. There is no one-size-fits-all industrial policy; countries can rely on a variety of instruments and institutions performing distinct functions more or less effectively, depending on the circumstances. (Andreoni, Chang, and Scazzieri, 2019).

And then, how about the effectiveness of industrial policy? UNIDO (2020) has mapped industrial policy instruments based on two criteria: replicability and effectiveness in the selected countries. The effectiveness of these industrial policy instruments will depend on the specific country context, the quality of its governance, and the package of interdependent policies that countries implement in order to climb the industrialization ladder. Moreover, there is no assurance that the success attained through best practice experiences can be replicated automatically.

However, there is one essential factor: government flexibility and capability in adapting programming and policy instruments are critical. Developing a government mentality that is not frightened of failure and encourages experimentation is even more crucial. Experiment can also be used to calibrate the appropriate level of selectivity of a policy instrument to the specific and evolving dynamics of a sector, technology, or market, which can result in demonstration projects. However, remember that frequent change can result in policy incoherence, increase business environment uncertainty, and undermine the long-term policy vision. Several G20 countries have put in place policies that work well and can be used by other countries. Some examples are:

- a. Extension service and vocational training to improve technology absorption, diffusion and adaptation (Best practices: China, Germany and Japan). According to Yang et al. (2023) German federal agencies have developed multiple strategies in response Vocational Education Training for industrial digitalization which is strengthening the capability of training digital talents through the modernization of the training regulations and framework curricula and promoting the digital transformation of VET through the financial support of projects and development of practical assistance.
- b. Management of natural resource rents to divert resources in productive development policies (Best practices: Indonesia and Malaysia). The government has restricted the export of raw materials from Indonesia, including nickel, in order to raise the added value of its natural resources. This restriction was made possible by the Regulation of the Minister of Trade of the

Republic of Indonesia Number 96/2019 on Export Provisions for Processed and Purified Mining Products as of December 30, 2019. The government mandates that foreign businesses establish plants in Indonesia for the downstream processing of nickel ore for export.

In conclusion, the government plays a crucial role in adapting its industrial policy to global demands and domestic conditions, particularly during the transition to sustainable industrialization.

III.3 Impact of Industrialization (Energy Consumption)

According to the International Energy Agency (2022), The industrial sector accounted for 38% or 169 EJ (Fig. 1) of the total world final energy demand in 2021, up from 33% in 2000. A persistent pattern of increased output in energy-intensive manufacturing subsectors has been a major driver of the growth in energy consumption during the previous decade. Due to this increase, industrial activity is now directly responsible for 9.4 GWh of CO₂ emissions, or 25% of the world's emissions. As a result of rising industrial product demand worldwide and more gradual improvements in energy efficiency, that amount of emissions has increased by more than 70% since 2000. Globally, the industrial sector has the highest ultimate electricity consumption. As of 2019, the industry accounts for approximately 42% of total electricity consumption.

According to the Ministry of Energy and Mineral Resources (2022), Indonesia's total energy consumption increased by 0.4% from the previous year to 909.24 million barrels of oil equivalent (BOE) in 2021. The industrial sector is the second-highest consumer of energy, accounting for 34.93 percent of national consumption.

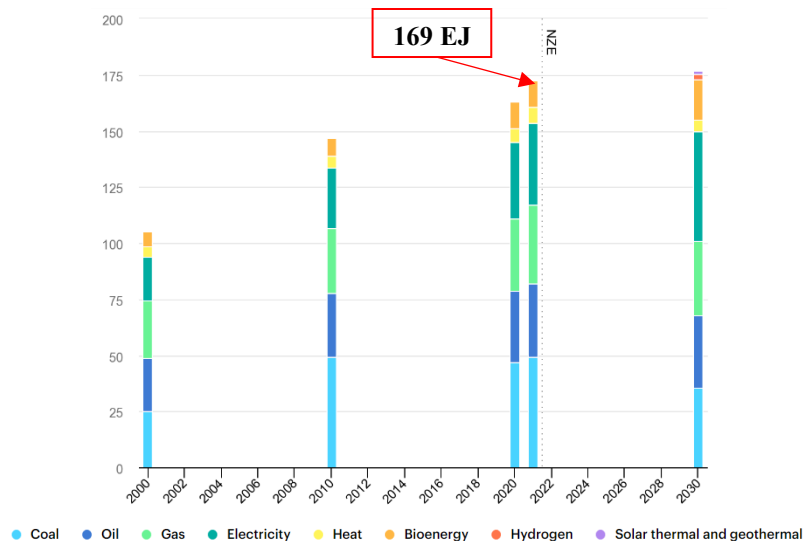


Figure 1. Final energy consumption for industry by fuel in the Net Zero Scenario, 2000-2030

(Source: IEA, 2022)

Based on these evidence, in order to meet the Net Zero Scenario goal, the industrial sector urgently has to transform to use renewable energy sources.

IV. Sustainable Industrialization

The term sustainable industrialization appears in Goal 9 (Industry, Innovation, and Infrastructure) of the United Nations Sustainable Development Goals, namely Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation. The goal of “sustainable industrialization” is to “promote inclusive and sustainable industrialization and, by 2030, significantly raise the industry’s share of employment and gross domestic product, in line with national circumstances, and double its share in least developed countries”. The OECD (2021) report identifies sectors whose primary business is related to SDGs. This methodology identifies economic activities that are related to the Sustainable Development Goals (SDGs) by measuring the value of these activities using discrete sectoral data for Canada, Denmark, Korea, Japan, and New Zealand and the SDGs 9 is the most linked to economic activities (Fig 2).

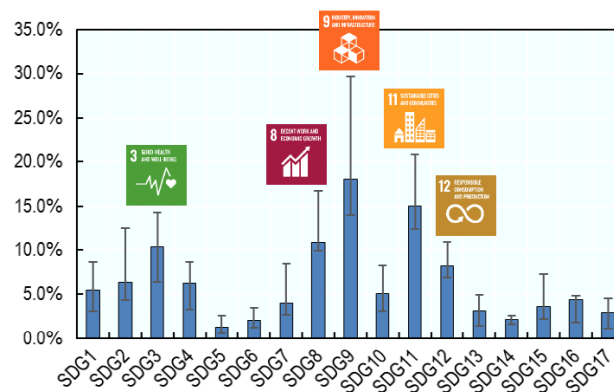


Figure 2. Economic activities linked to the SDGs, as a share of total value added, 2015

(Source: OECD, 2021)

According to Salonga (2021), sustainable industrialization refers to the transformation process towards a desirable industrialized economy characterized by wealth generation, rapid social development, and environmentally sound manufacturing all at the same time. Sustainable industrialization aims to protect scarce environmental resources while recovering economies. Kunkel and David (2021) emphasize that sustainable industrialization is the development and expansion of industries in a manner that minimizes negative environmental, social, and economic impacts while maximizing positive outcomes. It entails the adoption of sustainable production and consumption practices that enable industries to operate in an environmentally and socially responsible manner.

In order to meet a number of other SDGs, it could be claimed that sustainable industrialization requires separating economic growth from its effects on the environment, i.e. a decline in the use of resources and the emissions and waste that come from economic growth. Absolute decoupling can only happen if relative decoupling happens at least as fast as the economy grows. In other words, the amount of emissions per unit of resources needs to go down faster than the economy grows.

Consequently, stakeholders must take concrete measures to ensure that economic growth is accompanied by a drastic reduction in emissions. Government must enact the appropriate policies, and industry must transition to a more effective, efficient, and sustainable production process.

V. Sustainable Industrialization Policies Recommendation for Indonesia comparing with National Industry Development Master Plan of Indonesia

Indonesia has enacted Government Regulation No. 14/2015 regarding the National Industry Development Master Plan (Rencana Induk Pembangunan Industri Nasional, or RIPIN) for the period 2015–2035. In accordance with RPJMN 2015-2019, RIPIN is drafted to satisfy the requirements of Industry Law No. 3/2014 and to serve as a guide for government and industrial stakeholders. The government has since divided the Indonesian industrial development program for 2015–2035 into three phases.

1. The objective of the first phase of 2015-2019 is to increase the added value of natural resources in agricultural, mineral, and oil processing upstream industries, followed by the development of a supportive and dependable industry through the selective preparation of skilled and competent industrial human resources and the expansion of technological mastery.
2. The second phase of 2020-2024 will focus on achieving competitive advantage and environmental insight through the strengthening of industrial structure, technological mastery, and qualified personnel.
3. Phase three of 2025–2035 will be a time of establishing Indonesia as a powerful industrial nation with a robust and profound industrial structure, high global competitiveness, and innovation and technological foundation.

The government regulation was also intended to serve as a guide for the following national industry development strategies:

1. Developing the upstream and intermediate industry based on natural resources.
2. Controlling the export of raw materials and energy resources.
3. Improving the mastery of industrial technology and the quality of industrial human resources.
4. Establishing Industrial Development Region (Wilayah Pengembangan Industri/WPI).
5. Developing Central Region of Industrial Growth (Wilayah Pusat Pertumbuhan Industri/WPPI), industrial-designated regions, industrial estates and centres of small and medium industry.
6. Providing the affirmative action such as policy formulation, strengthening institutional capacity and providing facilities to small and medium industry.
7. Developing the industrial facilities.
8. Developing the green industry.
9. Developing the strategic industry.
10. Increasing the utilization of domestic products; and
11. Increasing the international industrial cooperation.

The Indonesian government has several industrial development objectives for the year 2035. They are listed below:

1. Two-digit growth of industry in 2035 to improve the contribution of industrial sector in GDP at 30%.

2. Improvement in domestic and international market-share by reducing the dependency to imported raw materials, auxiliary materials, and capital goods, as well as increasing the export of industrial products.
3. Acceleration of industrial deployment and distribution to all regions of Indonesia.
4. The increase of small industry contribution to national industrial growth.
5. The increase of innovation development and technology mastery.
6. The increase of employment rate of the industrial competent human resources; and
7. The strengthening of industrial structure with the growth of natural resources based upstream and intermediate industry.

These national industry development strategies and objectives appear to be an ideal plan. However, they are challenging to implement. There are several responses to National Industry Development Master Plan strategies and goals:

1. The high goals of the strategies, on the other hand, require cooperation from numerous stakeholders, since each strategy is run by a different technical ministry and coordination between ministries, state ministers, institutions, and sectors in Indonesia is a costly thing (because of ego sectoral).
2. The goal of industrial sectors is also not clear. Every sector, from mainstream to upstream, has a priority, which makes it hard to focus on industrial growth for a certain amount of time.
3. Current condition challenge:
 - a. Indonesia's supply chain flow and relationship between the upstream and downstream industries are still inadequate.
 - b. Negative deindustrialization: the deindustrialization process in Indonesia over the past few years was caused by a number disruption to the economic system, not by the natural effects of Indonesia's successful economic development. Middle-income dilemma is a warning to the Indonesian government about the danger of deindustrialization.
4. There are green industry standards and incentives (fiscal incentives) for green industry participants, but no national emission reduction objective. The Ministry of Industry has made little attempt to strengthen circular economy concepts. There are other additional Indonesian stakeholder laws that restrict the use of renewable energy, including one from the State Electricity Company that limits the use of solar panels.

This analysis has revealed that there is no policy in the current National Industry Development Master Plan that specifically addresses efforts to reduce emissions from the industrial sector. So, there are some policy recommendations for Ministry of Industry regarding sustainable industrialization (economic growth is accompanied by a drastic reduction in emissions) for the National Industry Development Master Plan of Indonesia:

1. Promoting carbon pricing for industry and environmentally related taxation

It is critical that the Ministries of Industry and Finance work together to implement a carbon tax on industries. The implementation of Indonesia's Law Number 7 of 2021 on Harmonization of Taxation (Including Carbon Tax) is scheduled for 2025. This rule must be put into effect right away, not later. According to IMF (2019), Carbon Pricing is an Effective and Low-Cost Strategy for Fighting Climate Change because governments can use the revenue to invest in clean energy and climate adaptation, stimulates clean technology, market innovation, and encourages investment in renewable energy. The OECD report emphasize that Carbon Tax and ETS are better than other policies like feed-in tariffs, rebates, and other regulatory instruments because the average cost of lowering a ton of carbon emissions can be up to eight times higher when other instruments are used.

2. Support Green Technologies in Industry

Governments can provide support for the development and adoption of green technologies, such as renewable energy and energy-efficient equipment for the Industrial sector to support the use of renewable energy and emission reduction goals. According to the Ministry of Energy and Mineral Resource or EMR (2022) Indonesia possesses a potential of 3,686 GW of new renewable energy derived from solar, wind, hydro, and geothermal sources; however, only 0.3%, or 11.5 GW, is utilized. This is a tremendous opportunity that Indonesia must exploit by developing green technologies. In the COP 26 meeting, the Minister of EMR emphasize that by 2025, the share of New and Renewable Energy (NRE) is targeted for 23% and will be dominated by solar PV and 57% in 2035 with the domination of solar, hydro, and geothermal power plants. The Minister committed to reducing Greenhouse Gas emissions by 29% by 2030 with our own efforts or 41% with international support.

3. Implementing the Circular Economy – Industrial Symbiosis Korean Model

Circular economy is the closed loop system in industrial estates allows industries to take advantage of each other to produce zero waste. Governments can promote the circular economy, which involves the reuse and recycling of materials and products. Industrial symbiosis in Korea is based on the same concept as circular economy, where waste from one company becomes raw material for other companies. This is strengthened by the existence of a legal foundation in the form of the Act to Promote Environmentally Friendly Industrial Structure (Legal Article 21). Korea has been implementing the concept since the first phase in 2005 in 5 industrial complex pilot projects until in 2019 it has succeeded in creating a national industrial symbiosis network for all its industrial complexes. According to Park (2022) the implementation of Industrial Symbiosis has resulted in significant economic results from 2007 to 2016: economic benefit by 2.2 billion USD; reduction in greenhouse gas emissions by 8,5 MT; and new investment by 0.73 Billion USD (Fig. 3).

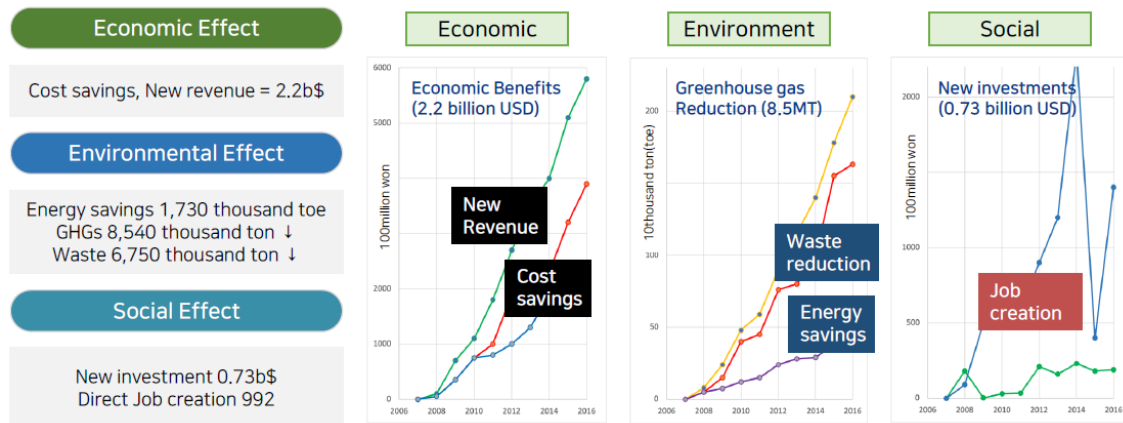


Figure 3. Industrial Symbiosis Performance in Korea 2007 - 2016

(Source: Park, 2022)

VI. Conclusion

The Indonesian G20 Presidency has concluded, and several significant decisions have been made regarding one of the three priority sectors, namely energy transition. Indonesia must establish a comprehensive national energy reform strategy that addresses every sector, including industry, as outlined in the leader's declaration. Industrialization plays a crucial role in a nation's economic development; according to Kaldor's first law, the growth rate of the economy (GDP) is positively correlated with the growth rate of the manufacturing industry sector. Industrial policies have been a significant driver of industrialization in all successful industrialization experiences. Therefore, industrial policies must adapt to context and time-specific country conditions, especially environmental change, because the industrial sector is the most consumed energy sector. To solve that problem, it is necessary to implement the sustainable industrialization concept in the nation's industrial policy.

This concept ensures that economic growth is accompanied by a drastic reduction in emissions. Indonesia has enacted Government Regulation No. 14/2015 regarding the National Industry Development Master Plan for the period 2015–2035. According to the strategies and targets, this national industry development appears to be an ideal plan but is challenging to implement because 1) it requires cooperation from numerous stakeholders since each strategy is run by a different technical ministry (ego sectoral); 2) the goal of industrial sectors is also not clear because every sector (from mainstream to upstream) has a priority; 3) Indonesia is facing negative deindustrialization; and 4) this plan does not contain the effort to reduce emissions and needs the adoption of the sustainable industrialization concept. To support the adoption of the sustainable industrialization concept in the National Industry Development Master Plan, there are some policy recommendations to be applied: 1) promoting carbon pricing for industry and environmentally related taxation: carbon pricing is an effective and low-cost strategy for fighting climate change; 2) support green technologies in industry: governments can provide support for the development and adoption of green technologies, such as renewable energy and energy-efficient equipment for the Industrial sector; 3) implementing the circular

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